Transfer Learning for Cross-Market Commodity Return Forecasting

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1 Introduction

1.1 Motivation & Background

Some of the more complex models seen in Section 2.1.1 suffer from needing of a lot of data. For some more niche markets with shorter histories and more constrained datasets this raises a problem. The most straightforward and perhaps naive solution is to simply train models on data-rich markets and implement them on data-sparse ones. However, it has been shown that traditional models fail when being transferred cross-market. When trying to transfer a convolutional neural network (CNN) model trained on US commodity futures to predict returns of the Chinese market, Hao et al. [1] find that this model fails to systematically outperform models trained on data specific to the Chinese market. These results suggest that there are little significant return-predictive patterns from US commodity futures which generalize across different markets. Hao et al. [1] give two explanations on why employing models cross-market fails in commodity markets. insert reasons Furthermore, Nguyen and Walther [2] have also found results to be heterogeneous across different commodity futures as no single model fit all commodities. It seems that traditional models lack cross-market transferability across both different countries and commodities.

1.2 The Research Gap

1.3 Research Questions & Hypotheses

1.4 Contributions to the Literature

1.5 Thesis Outline

2 Literature Review

This section will review the existing literature on the topic and work out the research gap this thesis aims to explore. Section 2.1 goes over existing approaches used in commodity forecasting and presents previous and remaining challenges. Section 2.4 explores the field of transfer learning as a potential solution for some of these remaining challenges. Section 2.5 shows where the existing literature leaves a gap that this thesis aims to fill, as the intersection of transfer learning, commodity markets and data-sparse emerging economies.

2.1 Commodity Forecasting

2.1.1 Classical Approaches

The discipline of commodity forecasting has gained more and more attention and approaches have gotten increasingly more safisticated.

2.1.2 Machine Learning & Deep Learning in Commodity Markets

2.2 Data Scarcity in Emerging Economies

2.3 Cross-Market Commodity Dynamics

2.4 Transfer Learning: Foundations & Applications

This shortcoming naturally invites the exploration of methodologies, which aim to built predictors, that allow for differences cross-market. The goal is to leverage data-rich markets, while still being able to include market-specific characteristics. This approach can be categorised as a transfer learning technique, which is a subset of machine learning, that allows for the assumption of different domains, tasks and distributions between the training and test set [3].

2.4.1 What is Transfer Learning?

2.4.2 Transfer Learning in Finance

2.4.3 Domain Adaptation Techniques

Mashetty et al. [4] explain approach This approach proved to outperform both a traditional machine learning model and a deep learning model, which were trained only on target data, in all three chosen metrics, namely mean absolute error (MAE), root mean squared error (RMSE) and directional accuracy.

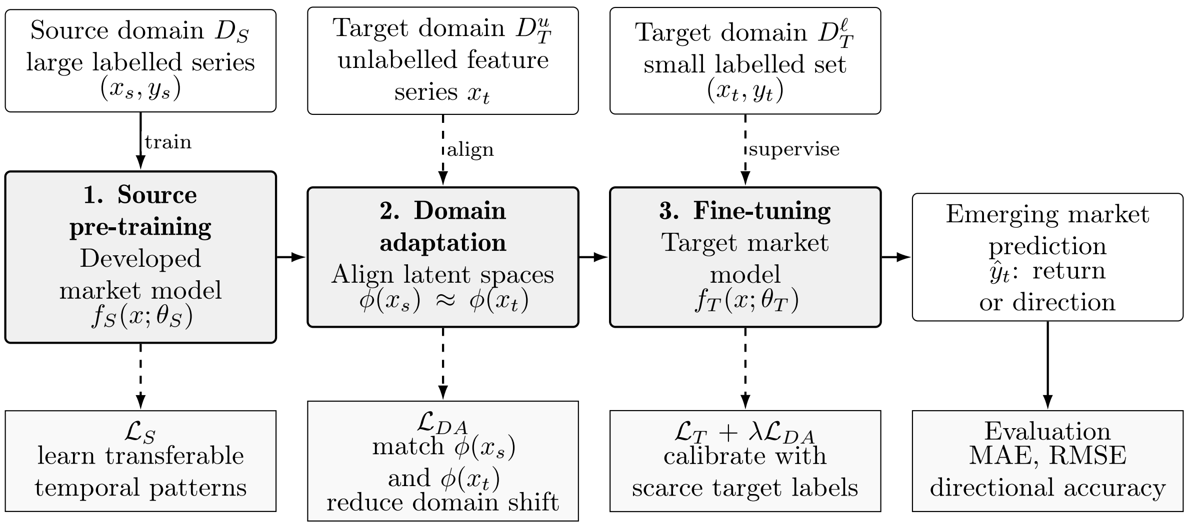
2.5 Research Gap & Positioning

3 Methodology

This section will build the transfer learning framework used to test whether forecasting performance can be improved compared to known approaches. Subsection 3.1 presents the three-stage pipeline, which mirrors the methodology of Mashetty et al. [4], and explains how the stages connect and what information flows between them. Subsections 3.2, 3.3 and 3.4 each explore one of the three stages: source pre-training, domain adaptation as well as fine-tuning and target prediction.

3.1 Framework Overview

Figure 1
Information flow in the three-stage approach



3.2 Stage A: Source Model Pre-Training

Let $P_s(t)$ be the price (futures settlement or spot) of the source commodity at time t and $y_s(t)$ the target as the log-return over horizon h with:

$$y_s(t) = \ln \frac{P_s(t+h)}{P_s(t)}. \tag{1}$$

Each input $x_s(t) \in \mathbb{R}^d$ at time t is a stacked vector of features computed over a lookback window w (so the model actually sees a sequence $\{x_s(t-w+1), ..., x_s(t)\}$ as input).

3.2.1 Model Architecture

3.2.2 Training Objective

3.3 Stage B: Domain Adaptation

3.3.1 Motivation for Distribution Alignment

3.3.2 Maximum Mean Discrepancy (MMD)

3.3.3 DANN Variant

3.4 Stage C: Fine-Tuning & Target Prediction

3.4.1 Fine-Tuning Strategies

3.4.2 Combined Loss & Optimisation

3.5 Baseline Models

3.6 Evaluation Metrics

3.7 Hyperparamenter Selection & Reproducibility

4 Data

This section will describe the empirical landscape of commodity markets and available data. Subsection 4.1 addresses the reasons for the specific choice of commodity classes examined. Subsection 4.2 and 4.3 describe the data used for the source and target market respectively.

4.1 Commodity Classification & Selection

4.2 Source Market: Global Benchmark Exchanges

4.3 Target Markets: Data-Sparse Economies

4.4 Feature Engineering

4.5 Descriptive Statistics & Stylised Facts

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5 Empirical Results

5.1 Main Results: Prediction Performance

5.2 Ablation Study: Decomposing Transfer Learning Gains

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5.4 Cross-Commodity Variation

5.5 Robustness Checks

5.5.1 Alternative Architectures

5.5.2 Alternative Source Markets

5.5.3 Sub-sample Stability

5.5.4 DANN vs. MMD

5.6 Economic Significance: Trading Strategy Simulation

6 Discussion

6.1 Interpretation of Core Findings

6.2 Comparison with Related Work

6.3 Economic and Policy Implications

6.4 Limitations

6.5 Future Research Directions

7 Conclusions

7.1 Summary of the Thesis

7.2 Answer to the Research Question

7.3 Broader Significance

7.4 Closing Remarks

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*not cited yet

List of Acronyms

CNN Convolutional Neural Network

MAE Mean Absolute Error

RMSE Root Mean Squared Error